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# 324514 (24)

BE (5th Semester)

Examination, Nov.-Dec., 2013

Branch : Electrical

## CONTROL SYSTEM ENGG.

*Time Allowed : Three Hours*

*Maximum Marks : 80*

*Minimum Pass Marks : 28*

**Note :** Answer all questions. Part (a) of each question is compulsory. Do any two from (b), (c) and (d) of each question. Do (b) or (c) of question IV.

**Q. 1.** (a) Write differences between open loop and closed loop systems. 2

(b) Write Mason's gain formula. Find overall transmittance using Mason's gain formula for Figure 1. 7



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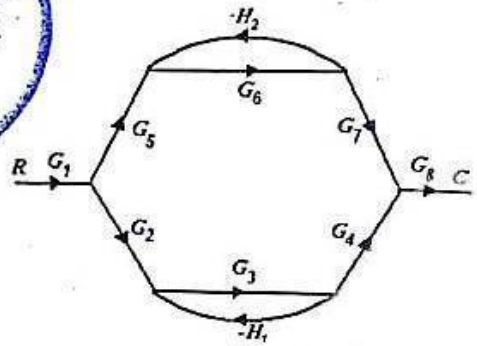


Figure-1

(c) Obtain analogous electrical system based on analogy and write mathematical equations describing the system given in Figure 2. 7

(i) F – V analogy

(ii) F – I analogy

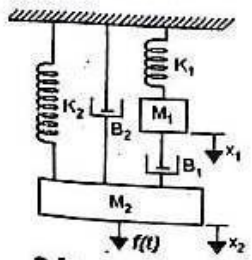


Figure-2



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(d) Apply block diagram reduction techniques and convert the block diagram given in Figure 3 into a single block.

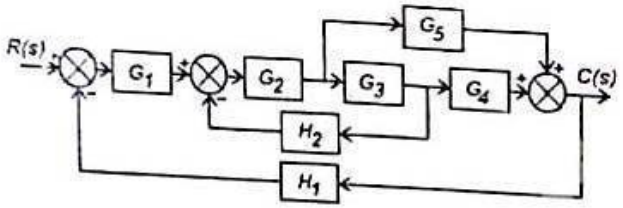


Figure-3

Q. 2. (a) Define "type" of a system. 2

(b) Draw the time response of a second order control system when subjected to unit step input. Derive expressions of time response specifications $t_r$, $t_p$, $M_p$ and $t_s$. 7

(c) A second order control system is represented by transfer function as : 7

$$\frac{X(s)}{F(s)} = \frac{1}{Js^2 + Bs + K}$$



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where F is the applied force and X is the displacement. Find J, B and K when a step input of 10N is applied and the test results are :

$$M_p = 5\%,\ t_p = 1 \text{ sec},\ c_{ss} = 0.2\text{m}$$

(d) Find for a unity feedback system having : 7

$$G(s) = \frac{10(s+2)}{s^2(s+4)(s+5)}$$

(i) Closed loop transfer function

(ii) Type of the system

(iii) $K_p$, $K_v$, $K_a$

(iv) $e_{ss}$ when input is $r(t) = 1 + 2t + 4t^2$

Q. 3. (a) What is gain margin and phase margin? Define gain cross over frequency and phase cross over frequency. 2



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(b) Draw the root locus plot showing all the steps for : 7

$$G(s)H(s) = \frac{K}{s(s+2)(s^2+8s+20)}$$

(c) Find the transfer function from the Bode plot given in Figure 4 and find $\omega_1$, $\omega_2$ and $\omega_3$. 7

(d) Draw the Nyquist plot and find the gain margin for the system : 7

$$G(s)H(s) = \frac{2.2}{s(s+1)(s^2+2s+2)}$$

Q. 4. (a) What is a compensator ? 2

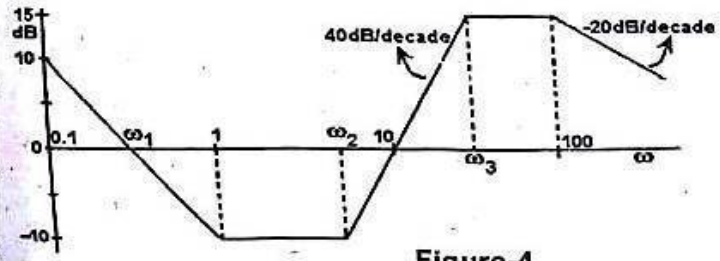


Figure-4

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(b) What are lead and lag compensators ? Draw the Bode plots on a semilog graph paper for a typical lead and lag compensator. Explain the effects and limitations of phase lag and phase lead compensators. 14

(c) Design a lead compensator with damping ratio $\xi = 0.5$ and $\omega_n = 2$ rad/sec for the system given by : 14

$$G(s)H(s) = \frac{K}{s(s+1)(s+4)}$$

Q. 5. (a) Define state, state variables and state vector. 2



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(b) Check controllability and observability of a system described by : 7

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -5 & -11 & -6 \end{bmatrix}, B = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, C = \begin{bmatrix} 10 & 5 & 1 \end{bmatrix}$$

(c) Represent the system in state space form described by : 7

$$4\frac{d^2x}{dt^2} - \frac{dx}{dt} + x(t) = 0$$

$$\frac{d^3x}{dt^3} + 5\frac{d^2x}{dt^2} + 2\frac{dx}{dt} = u(t)$$

taking $x_1(t) = x(t)$ and $x_2(t) = d^2x(t)/dt^2$ as the state variables and find the eigen values and eigen vectors of the matrix A. Also diagonalize matrix A.

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(d) Determine the state transition matrix output time response for the system

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$y = \begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$x_1(0) = 1; x_2(0) = 0$



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